INCH-POUND
MIL-PRF-32307
11 September 2008

PERFORMANCE SPECIFICATION

VALVE, QUARTER TURN, TRIPLE-OFFSET, TORQUE SEATED, SHIPBOARD USE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers triple-offset torque seated (TOTS), quarter turn valves for fuel oil, aviation fuel, lubricating oil, seawater, auxiliary seawater, freshwater, chilled water, and potable water piping system services. The associated hydraulic or electric gear operators/actuators are not covered by this specification.

1.2 Classification.

1.2.1 Types. TOTS valves are classified into two types based on materials of manufacture (see table I):

a. Type I – Non-ferrous

b. Type II – Ferrous

1.2.2 Body styles. The following TOTS valve body styles are covered by this specification. There are two styles (wafer and lug) and two thicknesses for each (short and long), totaling four style options. Short is equivalent to the face-to-face dimension equivalent of a MIL-V-22133 rubber-lined butterfly valve. Long is equivalent to the face-to-face dimension of a MIL-V-24624 high performance butterfly valve.

a. A – Short wafer

b. B – Long wafer

c. C – Short lugged wafer

d. D – Long lugged wafer

1.2.3 Seat style. Seating may be integral to the valve body or of removable design.

a. I – Integral

b. R – Removable

1.2.4 Sizes. TOTS valve sizes conform to the following nominal pipe sizes (NPS) in inches. The sizes listed in bold are most common and suggested for use whenever practicable:

Size (inches)	Size (inches)	Size (inches)
2½*	**4**	**8**
3	**5**	**10**
3½*	**6**	**12**

* - Less common valve sizes, use of which should be minimized for standardization and cost considerations.

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05M2, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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1.2.5 Flange styles. TOTS valves are to be designed for installation between the following flange styles:

a. ASME B16.5

b. NAVSHIPS Drawing 810-1385892

1.3 Part or identifying number (PIN). PINs to be used for TOTS valves acquired to this specification are created as follows:

M	**XX406**	-	**#**	**X**	**X**	**#**	**X**
Prefix for Military Specification	Specification Number		Type (see 1.2.1 & code below)	Body Style (see 1.2.2 & code below)	Seat Style (see 1.2.3 and code below)	Size (see 1.2.4 & code below)	Flange Style (see 1.2.5 & code below)

PIN Dash Number Explanation:

Type	PIN Code	Body Style	PIN Code	Seat Style	PIN Code	Size (Inches)	PIN Code	Flange Style	PIN Code
I	1	A	A	Integral	I	2½	025*	A	A
II	2	B	B	Removable	R	3	030	B	B
		C	C			3½	035*		
		D	D			4	040		
						5	050		
						6	060		
						8	080		
						10	100		
						12	120		

* - Less common valve sizes, use of which should be minimized for standardization and cost considerations.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATION

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections

QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500)

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-S-901	-	Shock Tests, H.I. (High Impact) Shipboard Machinery, Equipment and Systems, Requirements for
MIL-P-24396	-	Packing Material, Braided PTFE (Polytetraflouroethylene)
MIL-B-24480	-	Bronze, Nickel-Aluminum (UNS No. C95800) Castings for Seawater Service
MIL-V-24624	-	Valve, Butterfly, Wafer & Lug Style, Shipboard Service

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-798	-	Nondestructive Testing, Welding, Quality Control, Material Control, and Identification and HI-Shock Test Requirements for Piping System Components for Naval Shipboard Use

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

BUREAU OF MEDICINE AND SURGERY

BUMEDINST 6270.8	-	Procedures for Obtaining Health Hazard Assessments (HHAs)

(Copies of these documents are available from the Bureau of Medicine and Surgery, Department of the Navy, 2300 E Street, NW, Washington DC 20372-5300 or online at http://navymedicine.med.navy.mil.)

NAVAL SEA SYSTEMS COMMAND DRAWINGS

NAVSHIPS 810-1385892	-	Flanges, Bronze, Special for Use with Butterfly Valves

(Copies of these documents are available from the Commander, Naval Sea Systems Command, ATTN: SEA 05M2, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or by email at CommandStandards@navy.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN PETROLEUM INSTITUTE (API)

API-STD-598	-	Valve Inspection and Testing
API-STD-607	-	Fire Test for Soft-Seated Quarter-Turn Valves (DoD adopted)

(Copies of these documents are available from API Publications, Global Engineering Documents, 15 Inverness Way East, M/S C303B, Englewood, CO 80112-5776 or online at www.api.org.)

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ASME INTERNATIONAL

ASME B16.34	-	Valves − Flanged, Threaded, and Welding End (DoD adopted)
ASME B16.5	-	Pipe Flanges and Flanged Fittings NPS ½ through NPS 24 Metric/Inch Standard (DoD adopted)

(Copies of these documents are available from ASME International, 22 Law Drive, P.O. Box 2900, Fairfield, NJ 07007-2900 or online at www.asme.org.)

ASTM INTERNATIONAL

ASTM A182/A182M	-	Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service, Forged or Rolled Alloy-Steel (DoD adopted)
ASTM A240/A240M	-	Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications
ASTM A351/A351M	-	Castings, Austenitic, Austenitic-Ferritic (Duplex), for Pressure-Containing Parts (DoD adopted)
ASTM A479/A479M	-	Bars and Shapes, Stainless Steel, for Use in Boilers and Other Pressure Vessels (DoD adopted)
ASTM B148	-	Aluminum-Bronze Sand Casting (DoD adopted)
ASTM B150/B150M	-	Aluminum Bronze Rod, Bar, and Shapes
ASTM B865	-	Precipitation Hardening Nickel-Copper-Aluminum Alloy, Bar, Rod, Wire, Forgings and Forgings Stock
ASTM D512	-	Chloride Ion in Water (DoD adopted)
ASTM D1179	-	Fluoride Ion in Water (DoD adopted)
ASTM D1246	-	Bromide Ion in Water (DoD adopted)
ASTM D3223	-	Total Mercury in Water, Test Method for (DoD adopted)
ASTM D3557	-	Standard Test Methods for Cadmium in Water
ASTM D3559	-	Lead in Water (DoD adopted)
ASTM D4190	-	Standard Test Method for Elements in Water by Direct-Current Argon Plasma Atomic Emission Spectroscopy
ASTM D4327	-	Standard Test Method for Anions in Water by Chemically Suppressed Ion Chromatography
ASTM F467	-	Nuts, Nonferrous, for General Use (DoD adopted)
ASTM F468	-	Bolts, Nonferrous, Hex Cap Screws, and Studs for General Use (DoD adopted)

(Copies of these documents are available from ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.)

INSTRUMENTATION, SYSTEMS, AND AUTOMATION SOCIETY (ISA)

ANSI/ISA-75.02	-	Control Valve Capacity Test Procedures

(Copies of this document are available from the Instrumentation, Systems, and Automation Society, P.O. Box 3561, Durham, NC 27702-3561 or online at www.isa.org.)

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INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 3506-4	-	Mechanical Properties of Corrosion-Resistant Stainless-Steel Fasteners – Part 4: Tapping Screws
ISO/IEC 17025	-	General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of these documents are available from ISO, 1, rue de Varembé, CH-1211 Geneva 20, Switzerland or online at www.iso.org.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. TOTS valves furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list (see 4.2 and 6.4).

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3.2 Materials. TOTS valve assembly materials shall conform to Type I or Type II in accordance with table I.

TABLE I. TOTS valve assembly recommended materials [1/]

Description	Materials description			
	Type I (non-ferrous)		Type II (ferrous)	
Body and seat	ASTM B148 C95800	Ni-Al-Brz	ASTM A351/A351M Grade CF8M	Stainless steel
Integral seat	ASTM B148 C95800	Ni-Al-Brz	ASTM A351/A351M Grade CF8M	Stainless steel
Removable seat[2/]	ASTM B865 N05500 or ASTM B148 C95800	Ni-Cu-Al or Ni-Al-Brz	ASTM A240/A240M Type 316 ASTM A479/A479M Type 316	S31600
Seal ring[3/]	QQ-N-286 N05500 and flexible graphite	Composite; Ni-Cu-Al and Flexible Graphite[3/]	ASTM A182/A182M Type F316 ASTM A351/A351M Grade CF8M	Composite: 316SS and Flexible Graphite[3/]
Disc	ASTM B148 C95800	Ni-Al-Brz	ASTM A182/A182M Type F316 ASTM A351/A351M Grade CF8M	Stainless steel
Stem and stem-to-disc fasteners	ASTM B865 N05500	Ni-Cu-Al[4/]	ASTM A479/A479M Type XM 19	Stainless steel
Wetted fasteners	ASTM F468/467 Alloy 400/500	Ni-Cu or Ni-Cu-Al[4/]	ISO 3506-1 Grade A4	Stainless steel
Other wetted and moving parts	ASTM B148 C95800	Ni-Al-Brz	ASTM A351/A351M Grade CF8M	Stainless steel
	ASTM B150/B150M C63000	Ni-Al-Brz	ASTM A182/A182M Type F316 ASTM A351/A351M Grade CF8M	316SS
	ASTM B865 N05500	Ni-Cu-Al[4/]	ASTM A182/A182M Type F316 ASTM A351/A351M Grade CF8M	Stainless steel
Stem seals	MIL-P-24396	Braided PTFE	Braided Graphite and Graphite	
Brackets and adapters	MIL-B-24480 ASTM B148 C95800	Ni-Al-Brz	ASTM A182/A182M Type F316 ASTM A351/A351M Grade CF8M	Stainless steel
	Nickel-Copper QQ-N-281	Ni-Cu		

NOTES:

1/ Other materials may be substituted, but the burden of evidence and proof of equivalency or better must be provided by the supplier.

2/ Seawater service requires Ni-Cu-Al material.

3/ Other seal ring materials or designs may be used if shown to be equal or better and is acceptable to Naval Sea Systems Command. Burden of evidence shall be upon the supplier to provide evaluation criteria for compatibility with intended fluids, temperature and pressure.

4/ Ni-Cu in accordance with QQ-N-281 or ASTM F468/467 may be substituted where Ni-Cu-Al tensile strength is not required.

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3.2.1 Detrimental materials. Detrimental materials shall not exceed the maximum concentrations as specified in Table II.

TABLE II. Detrimental materials.

Material	Maximum concentration permitted (ppm)
Asbestos	None permitted
Halogens (total of chloride, fluoride and bromide) 1/	1,000
Lead	300
Cadmium plating	None permitted
Mercury and compounds (none intentionally added) 2/	10
Sulfur	10,000

NOTES:

1/ Materials containing more than 1,000 parts per million (ppm) total halogens are acceptable provided only fluorine exceeds 250 ppm and total leachable halides do not exceed 250 ppm.

2/ During manufacture, fabrication, handling, packaging, and packing, TOTS valve materials shall not come in contact with mercury or mercury compounds.

3.2.2 Toxins, carcinogens, reproductive hazards. The material shall have no adverse effect on the health of personnel when used for its intended purpose. The material shall be assessed by the Navy and Marine Corps Public Health Center using the administrative Health Hazard Assessment (HHA). A flowchart for this process can be found in BUMEDINST 6270.8. The HHA is a review of the material based on information submitted by the manufacturer to assess health hazards associated with the handling, application, use, and removal of the product. The material shall not cause any environmental problems during waste disposal.

3.2.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Design and construction. TOTS valves shall be of the types, body styles, seat styles, and sizes specified (see 1.2 and 6.2). The stem, disc, and seat shall have a double offset design. The third offset shall be achieved by inclining the seat and seal cone centerline with respect to the pipe and valve centerline in order to eliminate rubbing during 90 degrees of operation.

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3.3.1 Valve body. The dimensions of Type I and Type II valve bodies shall be in accordance with the requirements specified in ASME B16.34 for Class 150 valves. The minimum wall thicknesses for Type I and Type II valves shall be calculated in accordance with the basis equations for minimum wall thickness table of ASME B16.34 for Class 150 valves. Note that the design criteria specified in ASME B16.34 applies only to ferrous materials and must be modified to consider the stress limits for non-ferrous materials prior to use in the design of Type I (non-ferrous) valves. Valves shall be designed for installation between flanges in accordance with ASME B16.5 Class 150 for ferrous (Type II) valves and Drawing 810-1385892 for non-ferrous (Type I) valves. Wafer and lugged valves shall be manufactured with face-to-face dimensions in accordance with table III.

TABLE III. Valve face-to-face dimensions.

NPS (inches)	High performance butterfly equivalent MIL-V-24624 replacements (±0.0625 inches)	Rubber-lined butterfly equivalent MIL-V-22133 replacements (±0.0625 inches)
2.5	1.875	1.625
3	1.875	1.625
3.5	2.0	1.75
4	2.125	1.875
5	2.25	2.0
6	2.25	2.0
8	2.5	2.375
10	2.8125	2.563
12	3.1875	2.875

3.3.2 Actuator mounting bracket. The valve body shall have an integral provision for mounting the manual operator or motor actuator. An adaptive mounting bracket may be utilized to attach the actuator, provided it is approved by Naval Sea Systems Command and the entire assembly, valve, mounting bracket, and actuator, is successfully shock tested and vibration tested as specified (see 3.5.4 and 3.5.5).

Acquisition documentation for valve assemblies (see 6.2) shall identify requirements for mounting actuators including bolting size and pattern.

3.3.3 Valve seats. The valve seat shall be integral to, or removable from the valve body.

3.3.4 Disc. The seal ring assembly on the disc shall be replaceable without removing the stem and disc assembly.

3.3.5 Stem. The stem shall be of one-piece construction. The shaft shall be designed so that the failure of the shaft-to-disc connection or internal failure of the shaft shall not permit any portion of the shaft to be ejected from the valve as a result of internal pressure. The shaft seals shall be capable of being tightened while the valve is under pressure. The valve packing shall be made of a combination of die-formed and expanded PTFE or graphite and braided PTFE or graphite rings (see table I), or other sealing arrangement proven to be equivalent or better. Burden of proof of equivalency shall fall on the contractor and be submitted to Naval Sea Systems Command for approval (see 6.2).

3.3.6 Position indication. The indication of the position of the disc shall be integral with the valve shaft. The design shall ensure that the valve (and/or operator) cannot be assembled to falsely indicate the valve's open and closed position.

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3.3.7 Actuators. Enclosed worm gear type manual actuators shall be available for all size valves. The maximum allowable tangential force, based on hand-wheel size, is provided in table IV. This force applied to the manual actuator shall meet or exceed the maximum torque required to close and open the valve.

TABLE IV. Maximum tangential hand-wheel force.

Hand-wheel diameter (inches)	Total tangential force (rim of hand-wheel) (lbs)
2.5	30
3	30
3.5	30
4	30
5	50
6	72
8	102
10	120
12	135

3.4 Physical requirements.

3.4.1 Pressure-temperature ratings. The pressure-temperature rating for TOTS Type I valves shall be calculated in accordance with the non-mandatory method for Class 150 – all materials appendix of ASME B16.34. The pressure-temperature rating for TOTS Type II valves shall be in accordance with the "Ratings for Group 2.2 Materials" table in ASME B16.34 for Class 150, Standard Class valves. Both Type I and Type II TOTS valves shall be rated at 100 °F.

3.4.2 Seat tightness. TOTS valves shall meet the resilient seat leakage criteria as specified in API-STD-598 (see 4.5.9).

3.4.3 TOTS valve body marking. The following markings shall be permanently marked on the valve body:

a. Manufacturer name
b. Model number
c. Valve size in NPS
d. Valve pressure rating in psig

3.4.4 Identification plate. TOTS valves shall be provided with a permanently attached plate containing the following data:

a. Manufacturer name
b. Size of valve in NPS and pressure rating in psig
c. Body material
d. Valve trim material: stem, disc, seat
e. Contractor and Government Entity (CAGE) number
f. Component identification number (if known)

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3.5 Performance requirements.

3.5.1 Flow capacity. The minimum flow capacity shall be in accordance with table V.

TABLE V. Minimum flow coefficient (Cv).

NPS (inches)	Rated Cv*
2½	43
3	85
3½	114
4	180
5	371
6	732
8	1,472
10	2,480
12	3,790
* - Disc side direction flow coefficient	

3.5.2 Valve operation. TOTS valves shall be capable of fully opening from the fully closed position in 35 seconds, or less. TOTS valves shall also be capable of fully closing from the fully open position in 35 seconds, or less. Regardless of actuation method, TOTS valves shall operate under the pressure conditions and operating torque values listed in table VI.

TABLE VI. Maximum operating torque values.

NPS (inches)	Running (in-lb)	Pressure direction*	Max allowable to close (in-lb at 275 psig ΔP)	Max allowable to open (in-lb at 275 psig ΔP)
2½	320	SS	677	1120
		DS	677	1120
3	363	SS	1049	1720
		DS	1049	1720
3½	400	SS	1421	2320
		DS	1421	2320
4	451	SS	1793	2940
		DS	1793	2940
5	584	SS	3134	6162
		DS	3134	6162
6	451	SS	3492	6880
		DS	3492	6880
8	885	SS	4833	9534
		DS	4833	9534
10	1354	SS	8961	17614
		DS	8961	17614
12	1850	SS	13023	25626
		DS	13023	25626
*SS: Shaft Side Flow Direction *DS: Disc Side Flow Direction				

3.5.3 Fire resistance. TOTS valves, with actuators installed, shall be fire resistant and shall meet the allowable through valve and external leakage rate acceptance criteria when tested in accordance with API-STD-607 (see 4.5.3).

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3.5.4 Shock resistance. Valves and their actuators shall meet the Grade A shock acceptance requirements of MIL-S-901 when tested as specified in 4.5.4.

3.5.5 Vibration resistance. Valves and their actuators shall meet the Type I vibration requirements of MIL-STD-167-1 when tested as specified in 4.5.5.

3.5.6 Service life. TOTS valves shall be designed for a minimum service life of 30 years as demonstrated by acceptable leakage rates after 20,000 cycles (see 4.5.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall include the examination of 4.4 and the tests of 4.5.1 through 4.5.10 (see table VII) of the 4-inch, 8-inch, and 12-inch TOTS valves to qualify the TOTS valve sizes listed in 1.2.4. The 4-inch, 8-inch, and 12-inch sizes shall be used in lieu of the valve sizes specified in MIL-STD-798.

4.2.1 Extension of qualification. Tests conducted on automated valves, consisting of a valve and actuator combination, shall be extended to qualify the same actuator when installed on larger size valves. Successful passages of these tests for any body style wafer valve shall be extended to cover any body style lugged or wafer valve. Successful passage of these tests for long, double flanged design shall be extended to qualify short, double flanged design.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the tests of 4.5.7, 4.5.8, and 4.5.9 (see table VII).

TABLE VII. Examinations and inspections.

Test requirement	Requirement	Verification	Qualification	Conformance
Examination	3.1 through 3.3.7, 3.4.3, and 3.4.4	4.4	X	X
Cycle tests	3.5.2 and 3.5.6	4.5.2	X	
Fire test	3.5.3	4.5.3	X	
Shock test	3.5.4	4.5.4	X	
Vibration test	3.5.5	4.5.5	X	
Flow capacity test	3.5.1	4.5.6	X	
Functional check	3.5.2	4.5.7	X	X
Hydrostatic shell test	3.4.1	4.5.8	X	X
Seat tightness test	3.4.2	4.5.9	X	X
Detrimental materials	3.2.1	4.5.10	X	

4.4 Examination. Each TOTS quarter turn valve shall be examined for compliance with the requirements specified in table VII. Any redesign or modification of the manufacturer's standard product not qualified in accordance with 4.2 shall constitute rejection of the valves. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Methods of inspection.

4.5.1 Test valve sizes. TOTS valves of each of the 4-inch, 8-inch, and 12-inch sizes shall be subjected to the tests specified in 4.5.2 through 4.5.10. The 4-inch, 8-inch, and 12-inch sizes shall be used in lieu of the valve sizes specified in MIL-STD-798.

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4.5.2 Cycle tests. The cycle test shall be conducted using a test fixture that shall allow either side of the valve to be pressurized at the 100 °F rated pressure with ambient temperature, clean potable water when the valve is in the closed position. Type I valves shall close against a flow velocity of 15 feet per second minimum in the connecting pipe. Type II valves shall close against a flow velocity of 25 feet per second minimum in the connecting pipe. Once a side of the valve is pressurized, the valve shall be cycled to simulate operation in a shipboard system. A cycle consists of fully opening the valve from the fully closed position and returning it to the fully closed position. The valve shall undergo 20,000 cycles. Every 2,000 cycles the valve shall be tested for leakage in both the SS and DS directions using ambient temperature, clean potable water at the 100 °F rated pressure. The test pressure shall be held for 1 minute for valves up to, and including 6-inch NPS and for 2 minutes for valves 8-inch to 12-inch NPS valves inclusive. The number of cycles and the associated leakage rate shall be recorded. Every 2,000 cycles the flow direction shall be reversed to verify the bi-directional capabilities of the valve. During the cycle test, the stem seal may be adjusted. The operating time shall not exceed the requirements of 3.5.2. The torque to fully open the valve from the fully closed position and to fully close the valve from the fully open position shall not exceed the requirements of Table VI. The water leakage rate shall not exceed the maximum seat leakage rate of 10 cubic centimeters per inch of NPS per hour.

4.5.3 Fire test. TOTS valves shall be fire tested in accordance with API-STD-607. During the test, Type I valves shall be mated to the flat face flanges specified in 3.3.1. Type II valves shall be mated to ASME B16.5 Class 150 raised face flanges. Leakage shall be measured by monitoring the source tank water levels during the test. The total fire test period shall be the duration of the 30 minute fire test plus the time it takes the valve body's outer surface to cool to 212 °F. The fire test shall be conducted at a facility accredited by a nationally recognized organization to comply with ISO/IEC 17025. The facility shall demonstrate by past testing its ability to successfully conduct testing in accordance with API-STD-607 to the satisfaction of the U.S. Navy. An American Bureau of Shipping surveyor or U.S. Navy technical representative shall approve the vendor's selection of the test facility prior to conducting the test.

4.5.4 Shock resistance. Valves, with actuators installed (gear box, hydraulic, or electric), shall be tested in accordance with the Class I, Type A shock test requirements of MIL-S-901. For purposes of the shock test, the following MIL-S-901 classifications shall be applied to the TOTS valve:

a. Mounting location – deck or shell mounted as specified in 6.2

b. Mounting plane – in-line pipe

c. Mounting orientation – unrestricted

d. Test category – dependent on weight limitations for deck mounted valves; shell-mounted valves shall be heavyweight shock tested

During the shock test, the valve shall be pressurized at the 100 °F rated pressure with ambient temperature water. The valve shall be tested both open (normal operating mode) with pressure throughout and closed with pressure on the disc side only.

4.5.4.1 Initial post-shock test. Initial post-shock test shall consist of an external visual and dimensional check without disassembling the valve, a hydrostatic shell test in accordance with 4.5.8, and a seat tightness test in accordance with 4.5.9.

4.5.4.2 Final post-shock inspection. Tested valves shall be disassembled and inspected for breakage, cracking, deformation, and misalignment. Highly stressed areas and seating areas shall be checked for cracks using a dye penetrant test in accordance with MIL-STD-798. A dimensional check of the valve and valve internals shall be performed to determine valve damage. The valve and valve actuator shall meet the acceptance criteria of 3.5.4.

4.5.5 Vibration test. Valves, with actuators installed, shall meet Type I requirements of MIL-STD-167-1. Testing shall be conducted while the valve is pressurized at 100 °F rated pressure with ambient temperature water, once in the closed position, then in the open position. There shall be no evidence of valve structural damage or performance degradation.

4.5.6 Flow capacity test. Flow test shall be conducted in accordance with ANSI/ISA-75.02 to determine conformance with Table V. Flow capacities of the intermediate valve sizes may be extrapolated, provided the flow path geometry is a scaled version of the test valve.

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4.5.7 Functional check. TOTS valves, with actuators installed, shall be cycled (fully open-fully close-fully open) a minimum of three times while pressurized at the 100 °F rated pressure with ambient temperature water. The operating time and operating torque shall be measured to determine compliance with the requirements of 3.5.2.

4.5.8 Hydrostatic shell test. Each valve shall be subjected to a body strength and mechanical assembly test at 150 percent of the 100 °F rated pressure. The test shall be performed with water at a temperature not less than 60 °F and the valve shall be in the open position. The test duration shall be 1 minute for up to 6-inch NPS and 2 minutes for 8-inch, 10-inch, and 12-inch NPS. Any external leakage shall constitute failure of this test and shall be cause for rejection.

4.5.9 Seat tightness test. Following the body strength test, the valve shall be closed and the seat shall be tested for hydraulic and pneumatic tightness. The test duration shall be 1 minute up to 6-inch NPS and 2 minutes for 8-inch, 10-inch, and 12-inch NPS. The leakage rate shall not exceed the maximum leakage rate specified in API-STD-598 for resilient-seated valves when subjected to:

a. 110 percent of the 100 °F rated pressure, with ambient temperature water in the shaft side flow direction.

b. 90 psig pressure with air in the shaft side flow direction.

c. 110 percent of the 100 °F rated pressure, with ambient temperature water in the disc side flow direction.

d. 90 psig pressure with air in the disc side flow direction.

4.5.10 Detrimental materials.

4.5.10.1 Heavy metals. Heavy metals shall be analyzed in accordance with the methods specified in ASTM D3223, D3557, D3559, or D4190 to meet the requirements of 3.2.1. Specifications and certified material test reports for all valve materials shall be used to verify compliance with requirements.

4.5.10.2 Water leachable halides. Water leachable halides, the concentration of bromide, chloride, and fluoride ions in each filtrate shall be determined in accordance with ASTM D1246, D512, D1179, and D4327, as appropriate, and shall be corrected by the results of a blank determination (see 3.2.1). Specifications and certified material test reports for all valve materials shall be used to verify compliance with requirements.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. TOTS valves are intended for flow control in fuel oil, aviation fuel, lubricating oil, firemain, seawater, and freshwater systems installed on Navy surface ships and submarines. TOTS valves are not approved for services requiring Level I and SUBSAFE levels of essentiality.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title and date of this specification.

b. Quantity required.

c. Part or identifying numbers (PINs) required (see 1.2 and 1.3).

d. Applicable National Stock Number (NSN) if available.

e. Actuator to be supplied with the valve body: either (1) Manual operator or (2) motor actuator bolting size and pattern required (see 3.3.2 and 3.3.7).

f. Evidence for proof that the shaft will not be ejected in any circumstance is prints sufficiently detailed to portray the design consideration for shaft containment with an accompanying brief technical description. Format in order of preference is: PDF, AutoCAD, Microsoft Word (or equal), hard copy (8½ inches by 11 inches), Drawings of larger format, if necessary. All submissions should be emailed to commandstandards@navy.mil (see 3.3.5).

g. Shock test mounting location (deck or shell mounted) (see 4.5.4).

h. Packaging requirements (see 5.1).

i. Shelf-life requirements (see 6.5).

6.3 Definitions.

6.3.1 Disk side (DS) flow direction. Flow acting in a direction on a closed valve that would unseat the valve disc from the valve body (also known as the “non-preferred flow” direction, or “reverse flow direction” thru the TOTS valve).

6.3.2 Double-offset. Characterized by offset of the valve shaft behind the centerline of the valve sealing surface and an offset of the shaft to the side of the centerline of the pipe.

6.3.3 Stem side (SS) flow direction. Flow acting in a direction on a closed valve that would seat the valve disc on the valve body (also known as the “preferred” or “normal” flow thru the TOTS valve).

6.3.4 Triple-offset. Includes the double-offset design and incorporates a third offset which acts to create a camming motion, and effectively eliminates seat-to-seal rubbing throughout the valve’s 90-degree range of motion.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products, which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-XX406 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05M2, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to commandstandards@navy.mil.

6.5 Shelf-life. This specification covers items where shelf-life is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order. The shelf-life codes are contained in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoD 4140.27-M; Shelf-Life Management Manual, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points (ICPs), and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: http://www.shelflife.hq.dla.mil/.

6.6 Valve seats. Removable seating with material upgrade should be used for severe service such as firemain, seawater service, etc.

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6.7 Subject term (key word) listing.

Actuator

Fire resistant

Manually operated

Shutoff, bi-directional

Wafer, lugged

Custodians:
Army – AV
Navy – SH
Air Force – 99

Review Activities
Navy – AS, SA
Air Force – 03, 11
DLA – CC

Preparing Activity:
Navy – SH
(Project 4820-0928-000)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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